

No. of Functional Features Included in the Solution

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	5 Marks

Functional Features Overview

Lists every feature planned for the solution, its implementation status, and its contribution to the overall system — including the two features that were consciously scoped out, rather than presenting only the features that shipped.

S.No	Feature Name	Description	Module / Component	Status	Category
1	Dual-model emotion detection	Classifies problem text into 5 emotion classes using both BiLSTM and BERT.	src/inference/	Done	Core
2	Mixed-emotion detection	Detects when more than one emotion is plausibly present instead of forcing a single label.	src/inference/mixed_emotion.py	Done	Core
3	Trust-adaptive decision engine	Combines both models' outputs into one final emotion, trust level, and reason, based on real validation data.	src/orchestration/decision_engine.py	Done	Core
4	AI-generated response	Produces an empathetic, field-aware response via Gemini/OpenRouter.	src/generation/	Done	Core
5	Template fallback	Deterministic canned response when AI is off or fails, so the app never hangs on a broken/disabled LLM call.	src/generation/template_fallback.py	Done	Core
6	Model comparison / How I Decided	Optional expandable view of both models' raw predictions and the decision engine's reasoning.	src/ui/model_comparison.py, result_panel.py	Done	Supporting
7	Session history & CSV logging	Opt-in recording of every interaction to CSV, timestamped in IST.	session_history.py, csv_logger.py	Done	Supporting
8	Analytics dashboard	Emotion distribution pie chart, emotional-journey timeline, and per-field bar chart.	src/ui/analytics_dashboard.py	Done (2 of 3 tabs — Summary intentionally empty)	Supporting
9	CSV-based retrieval prediction	Predict from previously saved similar examples instead of re-running the models.	src/ui/csv_prediction_toggle.py	Pending — UI stub, disabled, no backing pipeline	Nice-to-have
10	Bring-your-own LLM API key	Let each user supply their own key instead of a shared free-tier one.	—	Dropped — would require refactoring the	Nice-to-have

S.No	Feature Name	Description	Module / Component	Status	Category
				LLM clients off their module-level singleton pattern	

Feature Summary

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	2 (both not implemented — one pending stub, one dropped)
Features Tested & Verified	8, covered by a 10-test Streamlit AppTest suite plus a manual click-through pass (see VALIDATION_REPORT.md)

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	5	Field/problem input form, settings toggles, decision & AI guidance panel, model comparison, analytics dashboard
2	Backend / Logic	3	Analysis pipeline orchestration, decision engine, mixed-emotion detection
3	Model / ML	2	BiLSTM predictor, BERT predictor
4	API / Integration	2	Gemini API client, OpenRouter API client
5	Persistence / Analytics	2	CSV interaction logger, session history + Plotly charts